

Sampling and Representation

Project: Transparent Book Recommender

1. Recall and choose your starting point

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Missing Data

Without opening the old lesson, explain one key idea from Missing Data. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Overfitting Intuition

Without opening the old lesson, explain one key idea from Overfitting Intuition. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Types of Classification Error

Without opening the old lesson, explain one key idea from Types of Classification Error. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

2. Practise with fading help

Use W2: Read the sample result for Questions populations and samples. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs · Procedure I chose · Middle steps · Result and check

Given inputs: _____

Procedure I chose: _____

Middle steps: _____

Result and check: _____

The step I completed without help, and the hint I needed

3. Individual reasoning check

Use the worked case for Questions populations and samples. Proposed explanation: “A successful run proves the conclusion”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

4. Independent transfer case

Independent case: Plan independently

Design an anonymous sample plan to ask all Grade 7 about three activity choices. State population, sampling frame, selection method, question, and how you will report nonresponse.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

5. Your learning target

A survey about school reading preferences includes only volunteers from a mystery club. Explain why 100 responses may still fail to represent the school. Propose a better sampling plan using fictional data.

6. Solve the given case

Only adventure-club members answer a school reading survey.

Explain the sampling problem.

Show your trace, calculation or evidence

Write or draw your reasoning and evidence.

7. Check a changed case

You survey only the chess club about all students' favourite books. Identify the sampling limitation and one improvement.

My independent answer and reason

Write or draw your reasoning and evidence.

Week 19 Test and reflect

8. Design a revealing test

Create a changed input, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record · My evidence

Test record: Changed input or condition

My evidence: _____

Test record: Expected answer or fallback

My evidence: _____

Test record: Observed result

My evidence: _____

Test record: Why the result occurred

My evidence: _____

9. Project checkpoint

Test reader profiles across reading levels, genres, ages, formats, and accessibility needs.

The project change and evidence I will keep

Write or draw your reasoning and evidence.

10. Self check

I can explain the mechanism. I can show evidence. I can handle a changed case. I can name a limit. Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

11. Transfer practice from the original programme

A survey about school reading preferences includes only volunteers from a mystery club. Explain why 100 responses may still fail to represent the school. Propose a better sampling plan using fictional data.

My independently reasoned answer

Record support used: none, hint or teacher explanation.

12. W1 Define the population

The question is whether all Grade 7 students prefer a longer lunch. Name the population and one unsuitable sample.

My reasoning and evidence

13. W2 Read the sample result

Seven of 12 fictional respondents prefer longer lunch. State the result without generalizing.

My reasoning and evidence

14. W3 Repair a claim

A report says all Grade 7 prefers coding after surveying only the coding club. Rewrite it.

My reasoning and evidence

15. W4 Plan independently

Design an anonymous sample plan to ask all Grade 7 about three activity choices. State population, sampling frame, selection method, question, and how you will report nonresponse.

My reasoning and evidence

16. W5 Separate invitation and response

Twenty students are selected fairly but only 8 answer. Does fair selection remove the need to report nonresponse?

My reasoning and evidence

17. Graphs and honest scales — W1 Read the scale

A bar is 6 squares long with 2 votes per square. How many votes is that?

My reasoning and evidence

18. Graphs and honest scales — W2 Compare fairly

Counts are 12 and 15. State the difference and ratio using the true values.

My reasoning and evidence

19. Graphs and honest scales — W3 Repair a misleading graph

Bars for 12 and 15 start at 10. Explain the misleading lengths and repair.

My reasoning and evidence

20. Graphs and honest scales — W4 Draw independently

Draw a bar chart for Music 6, Art 9, Coding 3. Include title, category labels, vote units, zero baseline and equal scale. State two supported comparisons.

My reasoning and evidence

21. Graphs and honest scales — W5 Limit the claim

Art has the most votes among 18 responses. Does that establish Art is best for learning?

My reasoning and evidence

22. Feedback, return check and learning record

Evidence · My record

Evidence: Worked alone / hint / model

My record: _____

Evidence: First step I repaired and why

My record: _____

Evidence: New case checked later; date and result

My record: _____

Evidence: What I can now explain without notes

My record: _____

Evidence: One remaining question

My record: _____

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

23. Pictorial case practice: Sampling and Representation

A fictional Grade 7 class wants to understand book-format preferences across its whole grade.

Population: all 60 Grade 7 learners. Frame: current school-approved list of those 60, used only by the teacher.

Teacher randomly invites 15; 12 answer the same neutral comic/prose/either question. No names enter the analysis table.

Sampling and Representation

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Define target

The question concerns all Grade 7, not only library-club members.



2. Select invitations

Teacher uses the complete approved list and documented random selection.



3. Record responses

12 replies from 15 invited; three nonresponses remain visible.



4. Report scope

Invitation response rate $12/15=80\%$; preference counts describe respondents.

CHANGE ONE THING

Invite only the 15 library-club members instead.

Explain the changed result using the matching lesson.

Can 80% response be reported as 80% of the whole grade responding?

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice evidence.

Project connection: Document whose requests inform the recommender and avoid treating an easily reached group as all intended users.

24. Project portfolio milestone

Project: Transparent Book Recommender

Current milestone: Evaluate and revise. Compare with your starting method, explain errors and record a justified revision.

Document whose requests inform the recommender and avoid treating an easily reached group as all intended users.

Evidence to keep · My record

Evidence to keep: This week's input and deciding step

My record: _____

Evidence to keep: Prediction and actual result (or labelled paper trace)

My record: _____

Evidence to keep: One change since the previous checkpoint and why

My record: _____

Evidence to keep: One error or limit and the next test

My record: _____

Explain what this week adds to your project. Reuse last week's record rather than starting again.

25. Supported practice and learning evidence

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A fictional Grade 7 class wants to understand book-format preferences across its whole grade.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Invitation response rate $12/15=80\%$; preference counts describe respondents.

1. Define target
2. Report scope
3. Record responses
4. Select invitations

My selected conclusion, deciding fact and explanation

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

After your attempt, compare with the supported explanation: Report scope Invitation response rate $12/15=80\%$; preference counts describe respondents. A sample plan states who could be selected, how selection happens and what happens after invitation. Random invitations do not guarantee that respondents resemble those who declined. Use a neutral question and preserve nonresponse coverage. Learners can discuss aggregate counts without receiving the identifiable selection frame.

Practice 2: explain the deciding evidence

Changed-case practice: Invite only the 15 library-club members instead. Which conclusion is supported by the supplied facts?

1. This one example guarantees correct results for every future input.
2. The invitation process no longer covers the whole grade through this frame.
3. We can decide the result without examining the changed input or the procedure.

My selected conclusion, deciding fact and explanation

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

After your attempt, compare with the supported explanation: The invitation process no longer covers the whole grade through this frame. Club members may differ in reading interests; selection is now limited to one group.

Repair a missing building block

Review Missing Data (ai-2/18). Goal: Detect absent and unusable values. Short practice: Why is the Boolean True rejected even though arithmetic can treat it like 1 in Python?. Return to this week and explain what you changed.

Review Overfitting Intuition (ai-2/16). Goal: Compare training and validation evidence for model capacity. Short practice: Does zero training error justify choosing Complex in the original record?. Return to this week and explain what you changed.

Review Types of Classification Error (ai-2/13). Goal: Build a labelled confusion matrix from matched reference and prediction pairs. Short practice: Which error causes a sound book to be sent for unnecessary repair review?. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual output or error, change one input, then run again. Keep code, predictions, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run evidence. Open the HTML copy to read or print it. Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test
